Maturation Makes Liars of Us All

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MATURATION MAKES LIARS OF US ALL

No man ever steps in the same river twice; for it is not the same river, and he is not the same man.

—HERACLITUS

This book is about how a group of men adapted themselves to life and adapted their lives to themselves. It is also about the Grant Study, now seventy-five years old, out of which this story came. In it I will offer tentative answers to some important questions: about adult development in general, about the people who engaged us in this exploratory venture, about the study itself, and, perhaps above all, about the pleasures and perils of very long scientific projects.

Originally the Grant Study was called the Harvard Longitudinal Study. A year later it became the Harvard Grant Study of Social Adjustments. In 1947 it received its now-official name, the Harvard Study of Adult Development. But to its members and its researchers, and in early books, it has always been the Grant Study.

It began in 1938 as an attempt to transcend medicine's usual preoccupation with pathology and learn something instead about optimum health and potential and the conditions that promote them. The first subjects were 64 carefully chosen sophomores from the all-male Harvard College classes of 1939, 1940, and 1941, who took part in an intensive battery of tests and interviews. That first group was joined by sophomores from the next three Harvard classes, resulting in a final cohort (as the panel of subjects in a study of this kind is called) of 268 men. The original intention was to follow these healthy and privileged men for fifteen or twenty years, supplementing the intake data from time to time with updates. Thus an abundance of information would accumulate about the men and the lives they constructed for themselves—information that could be analyzed at will over time and across different perspectives. (Interested readers will find much, much more on the history and structure of the Study in Chapter 3 .)

That plan was realized, and more. Almost seventy-five years later, the Grant Study still, remarkably, goes on. We're asking different questions now than the founders asked when the Study began, and our investigative tools are different. Of course the participants are no longer the college sophomores they once were; those who are still with us are very old men indeed. Time has called many of the beliefs of those days into question, and some much more recent ones, too. How long our current conclusions will hold up we cannot know.

But whatever the uncertainties, asking questions and trying to answer them is always a fruitful process. We actually have learned some of what they wanted to know back in 1938: who would make it to ninety, physically capable and mentally alert; who would build lasting and happy marriages; who would achieve conventional (or unconventional) career success. Best of all, we have seventy-five years' worth of data that we can refer back to (over and over again, if need be) as we try to understand why these things turned out as they did.

We can use those data to try to answer other questions, too. There are old ones still open from the early Study days—about the relative importance of nature and nurture, for instance, or how mental and physical illness can be predicted, or the relationship between personality and health. There are new ones that could never have been dreamed of in 1938, like what the emotions of intimacy look like in the brain. And some old questions have morphed into new ones as we learn to formulate ever more cogently just what we're seeking to accomplish in even making inquiries like these. This last in particular is an abiding concern of good science.

Thus this story of the Grant Study: how it came to be, how it developed, and how those of us who participated in it developed too. It's the story of what we've learned from it and what we haven't (yet). And it's a story about Time—studying it and living in it.

LONGITUDINAL STUDIES: THEIR STRUCTURE AND PURPOSE

The Grant Study is a longitudinal prospective study. Let me say a few words about that as we set out. In a longitudinal study, a group of participants (a cohort) is observed over time, and data about points of interest (variables) are collected at repeated intervals. In contrast, observations in a cross-sectional study are made only once. A longitudinal study can be prospective or retrospective. Retrospective studies look back over time, seeking to identify the variables that might have contributed to outcomes that are already known. Prospective studies follow a cohort in real time, tracking target variables as the subjects' lives proceed, and identifying outcomes only as they occur. Accordingly, the Grant Study collected all kinds of potentially (but not necessarily) relevant information about its cohort members over the many years of their lives, looking to learn what it could about health and success. And it has correlated this information periodically with the levels of health and success that each man actually achieved. You'll see many examples of this process as we go on.

Less abstractly, longitudinal studies let us contrast eighty-year-olds with themselves at twenty-five or sixty. Biographies and autobiographies, and the debriefings that elders offer their fascinated grandchildren, do the same thing. But these are all retrospective narratives, inevitably influenced by forgettings, embellishments, and biases. Time is a great deceiver.

Prospective studies are like the baby books compiled by doting parents, or like time-lapse photographs; they document change as it happens, allowing us to visualize the passage of time free from the distortions of memory. While butterflies recalling their youths tend to remember themselves as young butterflies, prospective studies capture the reality (hard to believe, and often avoided!) that butterflies and caterpillars are the same people.

But baby books and even year-long time-lapse videos are a dime a dozen. Prospective studies of entire lives are very rare indeed. None are known to exist before 1995; I'll have more to say on that shortly. In fact, observational data on adult development have been so sparse that when Gail Sheehy and Daniel Levinson produced their groundbreaking books on the subject in the 1970s, the cross-sectional studies they relied on led them to some very erroneous conclusions (such as the inevitability of the so-called midlife crisis). 1

Another important reason for prospective studies is that they establish contexts for the outcomes we're trying to understand. The most experienced handicapper in the world can't predict for sure which of the well-bred, handsome horses in the Churchill Downs paddocks each May will win the Derby. Certainty comes only after the race is won, and even then we only know who. The hows and whys remain a mystery. It's easy to rate a beautiful woman according to how many people think her so, or how well she conforms to some established ideal, or how long she keeps her looks. But ratings like that can't encompass the difference between being beautiful at one's high school prom and being beautiful at one's great-granddaughter's wedding, nor between the beauty at eighteen that is the luck of the draw, and the beauty at eighty that is the result of a life generously lived. The Grant Study was hoping to learn something about nuances like these in its exploration of success and "optimum" health.

Part of my intent in this book is to show why these nuances matter, and why, therefore, we need longitudinal studies if we want to learn about human lives. I will be focusing primarily on the Grant Study and its cohort, whom I will call the College men. But I will occasionally make reference to two other important cohorts. One is the Inner City cohort of the Glueck Study of Juvenile Delinquency. 2 The Glueck Study is a second Harvard-based prospective longitudinal lifetime study, begun independently of the Grant Study in 1940; its participants were a group of youths from disadvantaged urban Boston neighborhoods. Since 1970, one cohort of this group has been administered together with the original Grant Study, as part of the Harvard Study of Adult Development. (I will use that term when referring to both studies, but will continue to call the Grant Study per se by its original name.) I will also refer from time to time to the Terman women, a cohort from Stanford University's (1920–2011) Terman Study of gifted children, to whose data and participants I have had partial access. 3 It was the coming of age of the Terman cohort (male and female) in 1995 that made prospective lifetime data available for the first time. The Grant Study's access to the Inner City men and the Terman women permitted us to contrast our privileged, intelligent, and all-male College sample with another all-male sample of very different socioeconomic and intellectual profile, and with a group of even more intelligent, though not particularly privileged, women. When these contrasts are enlightening, I will note them. The Glueck and Terman studies (and some of the correlations among them) are described in their own contexts in Appendix B , and are also considered in my previous books Aging Well and The Natural History of Alcoholism Revisited. 4

I acknowledge readily that the Grant Study is not the only great prospective longitudinal lifetime study. There are others, three of which are better known than ours. Each has its own strengths and weaknesses. The Berkeley and Oakland Growth Studies (1930–2009) from the University of California at Berkeley include both sexes and began when the participants were younger; they provide more sophisticated childhood psychosocial data but little medical information. 5 These cohorts have been very intensively studied, but they are smaller and have suffered greater attrition than ours. The Framingham Study (1946 to the present) and the Nurses Study at the Harvard School of Public Health (1976 to the present) boast better physical health coverage, but they lack psychosocial data. 6 These are wonderful world-class studies, invaluable in their own ways, and more frequently cited than the Grant Study. But even in this august company the Grant Study is unmistakable and unique. It has been funded continuously for more than seventy years; it has had the highest number of contacts with its members and the lowest attrition rate of all; it has interviewed three generations of relatives; and, most crucial for adult development, it has consistently obtained objective information on both psychosocial and biomedical health. 7 Finally, perhaps alone among the world's significant longitudinal studies, the Grant Study has published, with the men's permission of course, lifespan histories as well as statistical data.

Other studies exist now in Great Britain, Germany, and the United States that are larger and more representative than these older ones, and will join them in length of follow-up in another decade or two. The Wisconsin Longitudinal Study, for example, began in 1957 and included about a third of all of Wisconsin's high school graduates of that year; it has endured for over half a century so far. 8 Eighty-eight percent of its surviving members are still active in the study at age sixty-five. (By way of comparison, 96 percent of the surviving Grant Study members are still active at age ninety!) The Wisconsin Study is more demographically representative than the other studies, and its economic and sociological data are richer and better analyzed. It has a weakness too, however; it lacks face-to-face medical examinations or interviews. We can anticipate a great wealth of prospective life data as these younger studies come into their own. But they will supplement, not supplant, the riches already offered by the Grant Study and its contemporaries.

When I came to the Grant Study in 1966, I was a very young man of thirty-two, and had not yet achieved the pragmatic relativism of Heraclitus. I had been studying stable remissions in schizophrenia and heroin addiction—recovery vs. no recovery, white vs. black. I spent my first ten years at the Grant Study identifying thirty unambiguously good outcomes and thirty unambiguously bad ones out of a randomly selected sample of one hundred middle-aged men from the classes of 1942–44. In 1977 I published a book, Adaptation to Life, demonstrating this accomplishment. 9 It made quite a splash. I was forty-three. What did I know?

Today I am seventy-eight. The men of the Grant Study are in their nineties. They're not the same as they were when they joined the Study, and neither am I. I have learned to appreciate how few blacks and whites there are in human lives, and how we and our rivers change from moment to moment. The world we live in is different; science is different; even the technology of documenting difference is different. As the Grant Study becomes one of the longest studies of adult development in the world, it is not only the men of six Harvard classes who are under the microscope. All of us who have worked on the Study—the Study itself, in fact—are now as much observed as observers.

ADULTS, GROWING

The laws of adult development are nowhere near as well known as the laws of the solar system or even the laws of child development, which were only discovered in the last century. It wasn't all that long ago that Jean Piaget and Benjamin Spock were still kids, and the phases of childhood so stunningly elucidated by them still regarded as unpredictable. Now, however, we watch children develop the way our ancestors watched the orderly waxings and wanings of the moon. We worry; we pray; we weep; we heave sighs of relief. But we are no longer particularly surprised. We know what to expect. Our libraries are full of studies of human development—up to the age of twenty-one.

What happens after that remains in many ways a mystery. Even the notion that adults do develop, that they don't reach some sort of permanent steady state at voting age, has been slow to gain traction. Some of the reasons for this are obvious. It is much easier to achieve a long perspective on childhood than on an entire life. And as in physics, once Time gets into the picture you can say good-bye to the old Newtonian verities. There are certainly patterns and rhythms to adult life, and when we circumvent the distorting effects of time upon our own vision, we can sometimes discern them.

But even the most carefully designed prospective study in the world can never free us completely from time's confounding influence. Lifetime studies have to last many, many years, and over those years everything will be changing—our questions, our techniques, our subjects, ourselves. I've been studying adult development since I was thirty, and I know now that many of my past conjectures, apparently accurate at the time, were contingent or just plain wrong. Still, the Grant Study is one of the first vantage points the world has ever had on which to stand and look prospectively at a man's life from eighteen to ninety. Having it doesn't save us from surprises, frustrations, and conundrums; quite the contrary. Nevertheless, a continuous view of a lifetime is now possible for human beings. Like that time-lapse film of a flower blooming that Disney made famous in the sixties, it is an awesome gift.

When the Study was undertaken in 1938, what we knew about human development across the whole of life was mostly based on inspiration or intuition. William Shakespeare delineated seven ages of man in As You Like It in 1599; Erik Erikson defined eight stages in Childhood and Society three hundred and fifty years later. 10 But Shakespeare and Erikson didn't have much by way of real data to go on. Neither did Sheehy and Levinson. Neither did I, in Adaptation to Life. Nobody had access to prospectively studied whole lifetimes. I hope that this book, written in 2012, will begin to correct that lack.

THE ART OF THE POSSIBLE

A few caveats as we proceed. It is well known that the Grant Study includes only white Harvard men; Arlen V. Bock, the physician who founded it, has frequently been criticized for arrogance and chauvinism on that account. It's less well known that the Grant Study was not an attempt to document average health over time, like the more famous Framingham Study, but to define the best health possible. And we must therefore keep in mind two realities. First, lifetime studies, like politics, are the art of the possible. As Samuel Johnson famously quipped about dogs walking on their hind legs, "It is not done well, but you are surprised to find it done at all." Second, in such huge undertakings, one must optimize one's chance of success. Columbia neuroscientist Eric Kandel did not choose a random sample of the world's population of homo sapiens when he did his Nobel-winning work on the biology of memory; he chose the obscure sea snail called Aplysia. Why? Because Aplysia has unusually large neurons. And it was precisely the gender and privilege of the Grant Study men that made them so useful for a study of human adaptation and development. Men don't change their names in midlife and disappear to follow-up as women do. Well-to-do men don't die early of malnutrition, infection, accident, or bad medical care, as happens much too often to poor ones. These men had a high likelihood of long life, a necessity for this sort of study. (A full 30 percent of the Grant Study men have made it to ninety, as opposed to the 3 to 5 percent expected of all white male Americans born around 1920.) Glass ceilings and racial prejudice were unlikely to hold them back from achieving to their fullest potential, or from the careers and lives that they desired. A Harvard diploma wouldn't hurt either. When things went wrong in the lives of the Grant Study men, they would have a good chance of being able to set them right. Last but not least, they were unusually articulate historians. Bock needed all those advantages. You can't study the development of delphiniums in Labrador or the Sahara. The Grant Study's College cohort and Aplysia may not be perfectly representative, but they both afford us windows onto landscapes we have never been able to see before.

One further note on Bock's choice of a homogeneous population. If we want to learn what people eat, we have to study many different populations. If we want to learn about gastrointestinal physiology, however, we try to keep variables like cultural habits and preferences uniform. Societies are forever changing, but biology mostly stays the same. This was another reason for the Grant Study's strategy. It was examining healthy digestion, not traditional menus. When possible, however, the Grant Study has checked its findings against other homogeneous studies of different populations, especially the disadvantaged men of the Inner City cohort and the highly educated women of the Terman cohort.

STUDYING THE STUDY

It is reasonable to ask whether this book is necessary. Over its seventy-five years of existence, the Study of Adult Development has so far produced 9 books and 150 articles, including quite a number of my own (see Appendix F). Why another? Well, because it's not always easy to see how reports of any given instant relate to the future, or even to the past. How do we understand a seventy-five-year-old article from a news daily? The short answer is, it depends on what's happened since. Whatever the world's papers were printing in the summer of 1940, England did not in fact fall to the Luftwaffe. Reporters do the best they can with what's available, but momentary glimpses can never capture the totality of history, and things may well look different later on. This in a nutshell is why longitudinal studies are so important. And to reap their full value, we have to take a longitudinal view of the studies themselves. When I sit down to summarize my forty-five years of involvement with the Grant Study, much of what I've written in the past seems to me as ephemeral as the 1948 headlines that trumpeted Dewey's victory over Truman, for only now do I understand how the story has really turned out. So far. As the people's philosopher Yogi Berra observed long ago, "It ain't over till it's over."

So as I see it, there are five reasons for this book. First, the Harvard Study of Adult Development is a unique, unprecedented, and extremely important study of human development. For that reason alone, its history deserves documentation.

Second, the Grant Study has been directed by four generations of scientists whose very different approaches beg for integration. The first director focused on physiology, the second on social psychology, the third (myself) on epidemiology and adaptation. The priorities of today's director, the fourth, are relationships and brain imaging. As any methodologist can see, the Grant Study had no overarching design. In 1938, there weren't enough prospective data on adult development even to build solid hypotheses. Like the Lewis and Clark Expedition and Darwin's passage on the Beagle, the Grant Study was not a clearly focused experiment but a voyage of discovery (or, as some have less charitably suggested, a fishing trip). The findings I report in this book are largely serendipitous.

This is the first time in forty-five years I have allowed this awareness into consciousness, let alone confessed it in print. In my repeated requests to the National Institutes of Health for funding, I always emphasized the potential of the Study—the power of its telescopic lens, its low attrition rate—rather than any specific hypotheses I planned to test. I analyzed data whenever I had a good idea; like a magpie I'd scope out our huge piles of accumulated material looking for a telltale glint or glimmer. I have often wondered whether there is a Ph.D. program in the country that would have accepted the plans of any of the four Grant Study directors as a thesis proposal. But what a wonderful harvest serendipity has produced. As I will demonstrate with pleasure, unpredictability is an inevitable and sometimes infuriating aspect of large prospective studies, but it gives them a startling richness that more narrowly focused endeavors can never achieve. All the better for magpies. Perhaps it was a good thing that I never took a course in psychology or sociology. At least I had no preconceived ideas ruining my eye for an emperor's new clothes.

Third, this book collects in one place material scattered across seventy years of specialty journals, in which the publications of each decade modified the findings of the previous ones, and were modified again in turn as new contexts cast new light on old data. Until now, for example, there's been no recognition of the importance of alcoholism in studies of development. But it's clear that earlier findings in some very unexpected areas are going to have to give way in the face of the Harvard Study of Adult Development's accumulating evidence about the developmental effects of alcohol abuse. It proved to be the most important predictor of a shortened lifespan, for one thing, and it was a huge factor in the Grant Study divorces, for another. Science is a changing river too.

Fourth, the Grant Study has seen and absorbed many theoretical and technological transitions, especially in the evolving field of psychobiology. It began in a day when blood was still typed as I, II, III, and IV. Race, body build, and (more speculatively) the Rorschach were considered potentially predictive of adult developmental outcomes. Data were tabulated by hand in huge ledgers—even punch cards sorted with ice picks were a yet-undreamed-of technology. For calculation, the slide rule ruled. Today we grapple with DNA analysis, fMRIs, and attachment theory, and 2,000 variables can be stored on my laptop and analyzed instantly as I fly between Cambridge and Los Angeles. Here, too, documentation and integration are called for.

A fundamental paradox is my fifth and final reason for writing this book. Despite all the changes I aim to document, we are all still the same people—the men who joined the Study seventy-plus years ago, and I, who came to direct it in 1966. One of the great lessons to emerge in the last thirty years of research on adult development is that the French adage is right: Plus ça change, plus c'est la même chose. People change, but they also stay the same. And the other way around.

A NOTE ON THE LIFE STORIES

I will be making my points in this book not only with numbers, but also with stories. The biographies of all living protagonists have been read and approved by them. All names are pseudonyms, but these narratives are not composites. They accurately depict the lives of real individuals, with the one stipulation that I have carefully altered some identifying details according to strict rules. I may substitute one research university for another, for example, but I do not substitute a research university for a small college, or vice versa—Williams College for Swarthmore, perhaps, but not for Yale. I allow Boston as an acceptable interchange for San Francisco, and Flint for Buffalo, but neither for Dubuque or Scarsdale.

I have made similar narrow replacements among types of illnesses and specific careers. In this way I have striven to remain faithful to the spirit of these lives and maintain their distinctive flavor, while altering the letter to ensure privacy. From time to time a prominent member of the Study has made a public comment relating to his participation in it. When I have quoted such comments, I have not disguised their authorship.

THE LIFE OF ADAM NEWMAN

Let me begin my history of the Grant Study with a story that illustrates many of the themes that have intrigued, enlightened, and confounded me over our years together. The protagonist is Adam Newman (a pseudonym), whose life—seemingly different with every telling—confronted us constantly with the realities of time, identity, memory, and change that are the heart of this book.

Newman grew up in a lower-middle-class family. His father was a bank worker who never finished high school. One grandfather had been a physician, the other a saloon owner. There was relatively little mental illness in Newman's family tree. Nevertheless, his childhood was grim. His mother told the Study that her first way of dealing with Adam's tantrums was to tie him to the bed with his father's suspenders. When that didn't work, she took to throwing a pail of cold water in his face. Later she spanked him, sometimes with a switch. Adam became extremely controlled in his behavior. He maintained a strict belief in and observance of Catholic teachings, and concentrated on getting all A's in school. His father was more lenient than his mother, but also more distant. "He only recognized that I was one of his children about once a month," Adam said. There was little show of affection in the family, and in Newman's 600-page record there is not a single happy childhood memory recounted. Rereading that record for this book, I see too that Newman says almost nothing about his father's death when he was seventeen.

In high school, Adam was a leader. He was a class officer in all four years, and also an Eagle Scout. He had many acquaintances, but no close friends. When he was a sophomore at Harvard, a few of his intake interviewers for the Study described him as "attractive," with "a delightful sense of humor." Others, however, described him as aloof, rigid, inflexible, repelling, self-centered, repressed, and selfish—the first of a lifetime of clues that this was a man of contradictions.

Newman's physical self was scrutinized minutely when he entered the Study, because the scientific vogue of the time was that constitutional and racial endowment could predict just about everything important in later life. He was described as a mesomorph of Nordic race with a masculine body build (all presumably excellent predictors for later success), but in poor physical condition. He was among the top 10 percent of the Grant Study men in general intelligence, and his grades were superior. As in high school, he had many acquaintances but few friends; he joined only the ornithology club and, eventually, that least social of fraternities, Phi Beta Kappa. One psychologist observed that he was "indifferent to fascism," and Clark Heath, the Study internist and director, noted that he "did not like to be too close to people."

In short, Adam lived mostly inside his own head. His personality was described as Well Integrated, but he was also considered a man of Sensitive Affect, Ideational, and Introspective; you'll hear more about all these traits as we go along. He gained distinction in the psychological testing on two counts: for his intellectual gifts, and for being "the most uncooperative student who ever agreed to cooperate in our experiments." In psychological "soundness" he ended up classified a "C," the worst category. (For much more on the assessment process, see Chapter 3 .)

The Study psychiatrists, informed by another theoretical fashion of the period, were more interested in Newman's masturbatory history than in his social life at college. They took pains to classify him as cerebrotonic, as opposed to viscerotonic or somatotonic. (These terms imply constitutional distinctions between people who live by the intellect, by the senses, and by physical vigor, but they were never satisfactorily defined.) It's also worth noting here, and it will come up again, that the early Study designers never put to the test their deep belief that body build is destiny. That had to wait for many years, even though opportunities for an empirical trial soon became available.

As a nineteen-year-old sophomore, Newman took a hard line on sex. He condemned masturbation, and he boasted to the Study psychiatrist that he would drop any friend who engaged in premarital intercourse. The psychiatrist noted in his turn, however, that although Adam disapproved of sexual activity, he was "frankly very much interested in it as a topic of thought." Newman told the psychiatrist a dream, too: two trees grew together, their trunks meeting at the top to form a chest with two drawers side by side, suggestive of a naked woman. He would wake from this dream, which visited him repeatedly, filled with anxiety.

To my mind, there was no adolescent in the Study who better exemplified psychoanalytic ideas about repressed sexuality than Adam Newman. And it was typical of the Grant Study's approach at the time that while Freud's theories were carefully explained to Newman (who vigorously rejected them), no one asked him about dating or friendships.

He was as dogmatic politically as he was sexually, and he regularly tore up the "propaganda" he received from the "sneaky" college Liberal Union. He claimed a commitment to empirical science both as an idea and as a career, but he also remained a practicing Catholic, attending Mass four times a week. When an interviewer wondered how his religious and scientific views fit together, he replied, "Religion is my private refuge. To attack it with my intellect would be to spoil it." More contradiction.

It wasn't until ten years later, when the scientific climate had shifted away from physical anthropology toward social psychology, and relationships had become a matter of interest, that the Study recorded that Newman had had few close friends in college besides his roommate, and that he had dated only rarely. In part this may have been because Adam was working to pay all of his college expenses himself, and was also sending money home to his fatherless family. But it's also true that he met early a Wellesley College math major who would become his wife and "best friend forever."

Adam went to medical school at Penn. He didn't want to minister to the sick, but he did want to study biostatistics and avoid the draft. He married during his second year there, but except for his wife he remained rather isolated. He had no more interest in World War II than he had in patient care. On graduation he fulfilled his military obligations with classified research at Edgewood Arsenal, America's research locus for biological warfare, and he continued in that work after the war. By 1950, Dr. Heath, the Study director, noted that Captain Newman, shaping the Cold War's nuclear deterrents but never seeing patients, was making more money than any of the other forty-five physicians in the Study. One of Newman's unclassified papers was "Burst Heights and Blast Damage from Atomic Bombs."

Despite these oddities, the record shows that by 1952, when he was thirty-two, Newman was steadily maturing. He was settling into a lifetime career in the biostatistics that he loved, using his leadership talents to build a smoothly running department of fifty people at NASA. His ethical concerns were engaged professionally too; in the 1960s his group participated in President Johnson's plan to put the military-industrial complex to work on the economic problems of the third world. His marriage would remain devoted for fifty years, by his wife's testimony as well as his own. It was an eccentric marriage—the two of them acknowledged each other as best friend, but neither had any other intimate friends at all. But many Study men with bleak childhoods sought marriages that could assuage old lonelinesses without imposing intolerable relational burdens, and Adam Newman seems to have found one. Perhaps this is why, unlike many men with childhoods like his, he effortlessly mastered the adult tasks of Intimacy, Career Consolidation, and Generativity (see Chapter 5) so early in his life.

Furthermore, whereas before he had denied intense feelings, perhaps out of fear of being overwhelmed, he was becoming more able to handle them. During the Study intake proceedings when he was nineteen, he described the atmosphere at home as "harmonious, my mother affectionate and my father the same." He was given a Jungian word association test (despite its preoccupation with physique, the Study did try, in an eclectic way, to measure the complete man), and Newman associated the word mother with affectionate, kind, prim, personal, instructive, and helpful. But the seasoned and tolerant social investigator who interviewed his mother experienced her as a "very tense, ungracious, disgruntled person," and Adam's sister commented, "Our mother could make anyone feel small." It wasn't until 1945, six years after he left home and five years after his initial workup, that Newman was able to write to the Study of his childhood, "Relations between me and my mother were miserable." "I don't remember ever being happy," he went on, and he recalled his mother having told him that she was sorry she had brought him into the world.

At age sixty, he took the sentence completion test developed by Jane Loevinger at Washington University to assess ego maturity. 11 The stem phrase was: When he thought of his mother . . . and Newman's response was . . . he vomited. Yet it would be an oversimplification to say that he went on to become steadily more conscious of difficult feelings, especially the painful relationship with his mother, because at age seventy-two he could not believe he had ever written any such answer. People are complicated; memory, emotion, and reality all have their own vicissitudes, and they interact in unpredictable ways. That's one of the reasons prospective data are so important.

The young Newman was extremely ambitious. “I have a drive—a terrible one,” he said of himself in college. “I’ve always had goals and ambitions that were beyond anything practical.” But by thirty-eight he had gained some insight into the “terrible” drive of the college years: “All my life I have had Mother’s dominance to battle against.” This realization caused a change in his philosophy of life. Now, he said, his goals were “no longer to be great at science, but to enjoy working with people and to be able to answer ‘yes’ to the question I ask myself each day, ‘Have you enjoyed life today?’ . . . In fact, I like myself and everyone else much more.” He hadn’t become a hippie; it was 1958 and that scene was still some years away, and in fact Newman’s fierce ambition was still burning in his heart. This was just another manifestation of the complexities of the man.

By the time he was forty-five, the laid-back freedom of his thirties was once more in abeyance as he battened down the hatches to deal with rebellious and sexually liberated daughters. Identifying unwittingly with his mother, he was now insisting that “greatness” be the goal of his intellectually gifted girls. Twenty years later, one daughter had still not recovered from his pressure. She described her father as an “extreme achievement perfectionist,” and wrote that her relationship with him was still too painful to think about. She felt that her father had “destroyed” her self-esteem, and asked that we never send her another questionnaire.

I wish I knew what happened between them in Newman’s later years, when he had mellowed remarkably. Alas, I do not. But I do know that Time continued to wreak its changes. Occasional backsliding notwithstanding, Newman became more flexible than he had once been. The tectonic shifts of the sixties, and being father to such adventurous progeny, loosened him up sexually. He stopped disavowing Freud’s theories. As his daughters reached adulthood, he (reluctantly) abandoned his prohibitions against premarital sexuality. He stopped fearing the “sneaky liberals” and began to share the view that law and order was “a repressive concept.” In fact, he came to believe that “the world’s poor are the responsibility of the world’s rich.” He quit his job in the military-industrial complex and put his scientific books under the house “to mildew.” At sixty he was solving agricultural problems in the Sudan, using statistical knowledge gleaned in the planning of retaliatory nuclear strikes. The man who had gone to Mass four times a week in college proclaimed, “God is dead, and man is very much alive and has a wonderful future.” In fact, as soon as Newman’s happy marriage had begun buffering the terrible loneliness of his childhood, his dependence on religion had diminished greatly. Eventually he became an atheist. In mid-adulthood his mystical side was being expressed in meditation, and he began teaching psychology and sociology at the university level.

This was not a complete changing of spots. To some degree, at least, we always remain the people that we were. On the one hand, Newman could write that he had learned from his daughters that “there was more to life than numbers, thought and logic.” But he still limited himself to only one intimate relationship—that is, with his wife. And although in his laboratory work he had become an increasingly generative leader, guiding those entrusted to his supervision with attention and concern, he was still a technician. Even in his teaching, he approached psychology and sociology not through the study of feelings, but through explorations of “the linguistic derivation of words like ‘relationship’ and ‘love.’ ”

So what had happened? Clearly these weren’t drastic personality changes. They weren’t intellectual changes, either. Newman hadn’t taken any riveting new psychology courses or met any charismatic psychiatrists. This was an evolution of his personality, which could be seen in the increased ability—not uncommon as people get older (see Chapter 5)—to stay aware of uncomfortable feelings without having to control or disown them. As Newman slowly became less anxious about his own sexuality, he had less need to condemn or control the sexuality of others. He was recapitulating a developmental process that is familiar to us in much younger people—the one that accompanies grammar-school children into the tumult of adolescence. As Newman struggled free of parental domination, he achieved a less constricted morality and became more comfortable with himself. In that greater comfort, he moved toward a greater comfort with, and willingness to be responsible for, others. None of the great psychologists of the nineteenth and early twentieth centuries, including Freud and William James, had had anything to say about adult maturational processes like this. But over the decades, we at the Grant Study have watched fascinated as Adam Newman and his fellows changed and grew.

Indeed, one great triumph of the Grant Study has been to make clinical documentation of changes like his available at exactly the moment when psychologists like Erik Erikson were beginning to recognize, schematize, and theorize about them. The loosening of the Sixties played a part in Newman’s evolution, but only a part. Not all of the Grant Study men did as he did at that time; some just dug their heels in deeper. This is one important truth of adult development: people grapple with growing up in their own ways, but we all do have to grapple. Obviously the men in the Study didn't all follow the same course of sexual development as Newman, any more than all children handle adolescence identically. But one way or another we all have to come to terms with our own sexuality, and the ways that we do (and don't) will end up shaping our lives.

Newman's story illustrates another endemic issue in long-term studies—the vicissitudes of subjects' memories. (It illustrates the vicissitudes of researchers' memories too, as you will see in a moment.) When I interviewed him at fifty, the only recurrent dream he could recall was of urinating secretly behind the garage, and he insisted that he had given up church as soon as he got to Harvard, having come to doubt the validity of religion. The four Masses a week and the dream of the naked woman/tree were as unavailable to him at fifty as his mother's unkindness had been when he was nineteen.

Newman's recollections of his history continued to alter as the years went on, always in the service of a process of psychological development; he was adjusting his inner self to the real world as he cautiously admitted more and more of his emotional life into consciousness. For example, when he was fifty-five, I asked Newman for permission to publish examples of his shifting memory. Repression again: he never became anxious about the fallibility of his beliefs; he simply returned my manuscript with the terse comment, "George, you must have sent this to the wrong person."

At sixty-seven, Newman told me that his approach to the rough spots in life was, "Forget it; let come what may." But as we might expect by now, that philosophy did not represent pure evolution from control freak to Zen-like detachment. On the contrary, in many ways he was returning to an adjustment much more like his earlier one. If no longer quite so buttoned-down as his mother had tried to make him, he had always been far more so than his unbuttoned daughters, and now he was buttoning up again. He abandoned the social intensity of teaching, and returned to the safety of numbers. From age fifty-five until he retired at sixty-eight he worked in city planning, managing the complexities of the new Texan megacities. Although he appeared at first glance to have had three widely varied careers—ballistic missile engineering, academic sociology, and this last enterprise—his creative expertise in multivariate statistics was the common denominator. And while he was flexible enough to become the office computer guru (despite being a generation older than most of the technological hotshots in the department), he never became Mr. Sociable.

"I don't know what the word 'friend' means," he said at seventy. He gave up meditation. "I ruined some of its delights by reading too much neuropsychology," he told me at seventy-two, in a comment reminiscent of his early fear that intellectual scrutiny would destroy his Catholicism. Still, his passion at the time was nuclear disarmament—an attempt to undo the aggressive achievements of his military youth. It was also an example of the developmental stage of Guardianship, which we'll explore in Chapter 5 . And one of his bulwarks always was his gift for humor, a characteristic not shared by all statistical types. Even as a young man, he could take some distance from the complaint that he had "twice the sex drive" of his wife, and give his grumbling a wry and graceful turn: "We believe that making love should be practiced as an art form."

In our last interview, Adam Newman still maintained that he knew nothing about friendship, yet he loved his wife from his teens to the end of his days. His plan for the dread possibility that she might die before him was to join the Sierra Club and hang out with "tree huggers," yet he was happiest working with numbers. He avoided too much contact with people, but during our interview the birds came to eat seeds literally from his hands, which he lovingly held out for them. He had abandoned "spirituality" years before and had not set foot in a church for years before that, but he showed me proudly the awe-inspiring fractals he was creating on his computer. He was a bundle of superficial contradictions, but he wasn't alone in that. Everyone is consistent in some things and not in others, yet ultimately true to some fundamental essence in themselves. The more things change, the more they stay the same; Newman was part mystic and part engineer, and he remained that way to the end.

He remained true to repression as a defense mechanism, too. When he was seventy-two I asked him for the third time if he could remember any recurrent dreams from childhood. This time he didn't miss a beat: "Do you mean the one when I was on roller skates and ran into the back porch?" Three times over thirty years he is asked for a recurrent dream from his adolescence; each time he comes up with a completely different one with no recollection of the others. He relied on repression at nineteen, and at seventy-two he still did.

But that doesn't mean he hadn't changed. He wasn't living purely inside his head anymore; the narcissism of his youth had given way more and more over the years to interest and to empathy. And his mood was light. A very unhappy college student had become a very contented old man. As I closed our interview, I asked him if he had any questions about the Study to which he had contributed for more than fifty years. He had one. "Are you having a good time?" As I prepared to take my leave I politely shook his hand and he—so uncooperative and self-centered in his college days, so shy and intellectual for the last two hours—exclaimed, "Let me give you a Texan good-bye!" and threw open his arms to give me a huge hug.

Reviewing my write-up of that interview eighteen years later for this book, I found that I had closed with the simple words, "I was entranced." But I also found, to my chagrin, that my own memory was as fallible and as defensive as his. I had been remembering the fifty-five-year-old man who appeared in my first book; I had expected to be relating the tale of an aging hippie. I had completely lost track of his return to city planning (which is, after all, engineering, if of a gentler form than bomb design). One lesson for me in the writing of this book has been that retrospection is as unreliable in investigators as in their subjects. Without prospectively written records it is very easy to repress unpleasant truths, and I was as good at it as Newman was. Even rereading my own transcript wasn't enough to keep me honest; that took an encounter with his death certificate. For twenty years I had conveniently forgotten that at the time of that interview, when he was seventy-two, he was a man preparing to die of an ugly and debilitating cancer. Out of my own dread of death, I couldn't keep his impending demise in mind. How could I claim any understanding of what was going on in him while failing to take that into account?

Be that as it may, less than a year before his life was consumed by the insatiable greed of his malignancy, Adam Newman wrote his last words to the Study: "I am happy." He had become a model of Erikson's final life stage, the one he calls Integrity (Chapter 5). This was an achievement that was still beyond me; Newman was not nearly as scared of his dying as I was, which probably had a fair amount to do with my failure to recall his cancer. My own father had died too soon to teach me much about adult life, but remembering Newman's life reminded me of how much I have learned from the Grant Study and its members in his stead. The dying can be happy too.

CONCLUSION

As a child I had little interest in microscopes, but I admired the telescope on Mount Palomar—then the most powerful one in the world. I wanted to see forests in their entirety, not just trees. Later on, as a psychiatrist, I longed to see beyond mere snatches of life to the whole picture. In this book I will focus on the rich potential of a telescopic view of life. There will be plenty of surprises, because a long enough view can turn conventional views of causation upside down. For instance, studied prospectively, physical health turns out to be just as important a cause of warm social supports and vigorous exercise as exercise and social supports are causes of physical health. Some readers will surely be outraged at such heresy, but as Galileo discovered, telescopes can get people into a lot of trouble. Long-term studies are as unsettling as they are enlightening.

To add to the uncertainty, we don't know how far to trust even our latest findings. Time changes everything, and it makes no exceptions for longitudinal studies. It transforms the world we live in while we're living in it, and pulls scientific thinking forward even while making it obsolete. None of this can be helped; it's an intrinsic hazard of long endeavors. The more powerful the telescope, the more likely it is that the light we are seeing through it is many thousands of years old. The Grant Study is only seventy-five, but that's more than threescore years and ten, and in the context of a man's life, a very long time. Many of the early findings of the Study are ill-conceived, out-of-date, and parochial; some of our later findings will likely prove to be so too. But some, I hope, will endure. And in the meantime, they give those of us who are curious about our own lives, and the lives of those we cherish, plenty to think about.

It reminds me of my first day of medical school. "Boys," the Dean told us (this was in 1955), "the bad news is that half of what we teach you will in time be proven wrong; and worse yet, we don't know which half." Still, half a century later, our class has done pretty well by its patients. So I maintain hope that the old-fashioned Grant Study, twentieth-century artifact though it be, can offer some fresh wisdom and some real inspiration to twenty-first-century readers.